

# Fauzan Mohammed

mohammedfauzan44@gmail.com | linkedin.com/in/fauzanmohammed

## SUMMARY

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Experienced Data Scientist with expertise in NLP and LLM systems, adept at designing and deploying AI solutions at scale. Proven track record in building production guardrail infrastructure and prompt engineering, ready to contribute to Pinterest's **Trust & Safety** initiatives.

## EDUCATION

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**Rensselaer Polytechnic Institute**

*Masters in IT/CS*

Troy, NY

*Aug 2023 – Dec 2024*

**Kwame Nkrumah University of Science and Technology**

*Bachelor of Science in Mechanical Engineering*

Kumasi, Ghana

*Sep 2016 – June 2020*

## EXPERIENCE

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**Data Scientist – NLP/LLM Systems + Responsible AI**

*Walmart Global Tech*

Aug 2024 – Present

*Sunnyvale, CA*

- Contributed to the design and deployment of a production multi-agent LLM system powering Walmart's conversational shopping platform, orchestrating intelligent search, guided discovery, and personalized recommendations for 20M+ monthly active users.
- Designed prompt-tuned LLM-as-judge evaluation frameworks for search quality and product relevance assessment across raw and reranked results, driving measurable improvements in ranking precision across product categories.
- Built production guardrail infrastructure using prompt-tuned LLM judges on GCP and Element Gateway, implementing real-time safety enforcement and hallucination detection to ensure responsible AI compliance for 10M+ users.
- Designed and maintained prompt templates, multi-step chains, and persona configurations across agent workflows, implementing version-controlled prompt libraries for consistency and reuse across the conversational platform.
- Developed an AI agent-based Event Title Generator leveraging LLM function calling to trigger internal APIs, autonomously infer user intent, map to structured event categories, and generate personalized titles within the shopping experience.
- Architected multi-turn conversation management logic including memory handling, contextual continuity, and dynamic persona assignment across domain-specific agent roles.
- Developed unsupervised NLP pipeline using DBSCAN clustering for topic modeling and intent extraction, expanding chatbot query coverage from 15% to 74%.

**Data Scientist, Intern**

*Walmart Global Tech*

May 2024 – Aug 2024

*Sunnyvale, CA*

- Engineered Transformer-based fallback detection system using semantic clustering to classify 100K+ daily fallback queries into 27 intent clusters, improving diagnostic accuracy by 6%.
- Optimized LLM router architecture through prompt engineering and query decomposition, improving task-selection accuracy by 4% and reducing latency by 18%.
- Built automated evaluation dashboards using SQL and **Python** for real-time monitoring of precision, recall, and prompt drift, reducing experimentation cycles by 30%.

**Junior Machine Learning Engineer**

*Omdena Thrive*

Dec 2022 – Apr 2023

*Remote*

- Engineered BERT-based NLP models with attention mechanisms achieving 85% classification accuracy on 5K+ mental health journal entries; deployed production analytics dashboard on GCP using Flask and Plotly for 200+ clinical users.

PROJECTS

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AI-Powered Financial Rule Automation System

Dec 2024

- Built AI rule-generation engine integrating multiple Transformer models (MiniLM, RoBERTa, DeBERTa) with QuickBooks API, implementing multi-task fusion architecture to auto-categorize transactions at 92% precision and reduce manual rule creation by 70%.

LEADERSHIP

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Co-founder and Director

*Fama Foundation*

Aug 2019 – Present

*Accra, Ghana*

- Led impactful programs across 10 Ghanaian communities, enrolling 325 teenage girls in computer programming and literacy education.

TECHNICAL SKILLS

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**Languages:** Python, SQL, Java

**ML/DL:** PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, Sentence Transformers, spaCy, NLTK

**LLM & AI:** LangChain, LlamaIndex, FAISS, OpenAI/Anthropic/Gemini APIs

**Cloud & Data:** GCP (Vertex AI, BigQuery, Cloud Run), Docker, Kubernetes, Pandas, NumPy, Spark

**MLOps:** MLflow, Weights & Biases, Flask, FastAPI, CI/CD, Git, DVC

**Specialties:** Multi-Agent Systems, LLM Fine-Tuning (SFT, RLHF, DPO), Prompt Engineering, RAG, Model Evaluation, Responsible AI, Function Calling & Tool Use, Conversation Flow Design